12 – Submodular Maximization (II)

IEOR 145/245

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Project Teams

- Please use the excel sheet to **reach out to each other**

- Project teams **due via bCourses Today**



Greedy Algorithm

$$\max_{S \subseteq N, |S| \leq k} f(S) = \max_{S \subseteq N, |S| = k} f(S)$$

- Idea: Pick best but *accounting for past steps*
- Define marginal value function,

$$\Delta f(e|S) = f(S + \{e\}) - f(S)$$

1. **Build solution in k steps**
2. **Always add element with maximum marginal value to current set**

Add $\text{argmax}_{e \in N \setminus S} \Delta f(e|S)$

How “Good” is Greedy?

- We know it is not optimal
  - Example from probability maximization last lecture
- How “bad” can it be?
  - Need a metric to measure worst case performance

Approximation Guarantee
[A Metric for "How Good?"]

- Let ALG be an **efficient** heuristic for the problem
  - Such as Greedy

- Let OPT be the (intractable to compute) optimal

- **Approximation guarantee** of ALG is:

$$\min_{I} \frac{\mathrm{ALG}(I)}{\mathrm{OPT}(I)} \qquad (I \text{ is problem instance})$$

Difference from competitive ratio: *gap due to computation, not due to uncertainty*

Greedy: Best Possible Guarantee

- Greedy is $\left(1 - \frac{1}{e}\right) \approx 0.63$ approximate when f is monotone and submodular
  - i.e., Greedy always outputs a solution with value at least 63% of OPT
  - True for probability maximization too

- **No efficient algorithm can beat this**

- **Performs exceedingly well in practice**
  - >90% of "optimal" in many applications

Submodular Maximization with Budget Constraint

Budget Constrained Optimization

Let $b(S) = \sum_{e \in S} b_e$,

$$\max_{S \subseteq N, b(S) \leq B} f(S)$$

- Add greedily while there is budget remaining?

Budget Constrained Optimization

Let $b(S) = \sum_{e \in S} b_e$,

$$\max_{S \subseteq N, b(S) \leq B} f(S)$$

- Add greedily while there is budget remaining?
- **Can have very bad performance**
  - $f(\{1\}) = 10 \quad \& \quad f(\{e\}) = 9 \; \forall \; e \in \{2,3,4, \ldots .10\}$
  - Budget of 9 and $b_1 = 9 \; \& \; b_e = 1 \; \forall \; e \in \{2, \ldots 10\}$
  - Greedy will just pick element 1, but optimal is 8 times as much

Value-per-unit

- Hiring with a budget of 9:

  one “star” costs 9 and brings value 10;

  nine juniors cost 1 each and bring 9 each

- Plain greedy grabs the star (10 > 9) and uses up budget

- Greedy on value per unit of budget fixes the issue:
  - Star: $10/9 \approx 1.1$ per unit; each junior: $9/1 = 9$ per unit → juniors first
  - Like comparing unit prices at the grocery store

Budget Constrained Optimization

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  - Budget of 9 and $b_1 = 9 \; \& \; b_e = 1 \; \forall \; e \in \{2, \ldots 10\}$
  - Greedy will just pick element 1, but optimal is 8 times as much

- Greedy w.r.t. "value-per-unit-budget"

Add feasible element $e = \text{argmax}_{e \in N \setminus S,\, \boldsymbol{b(S)+b_e \leq B}} \; \dfrac{\boldsymbol{\Delta f(e|S)}}{\boldsymbol{b_e}}$

Example

Add feasible element $e = \text{argmax}_{e \in N \setminus S,\, \boldsymbol{b}(\boldsymbol{S}) + \boldsymbol{b_e} \leq \boldsymbol{B}} \frac{\boldsymbol{\Delta f(e|S)}}{\boldsymbol{b_e}}$

- $N = \{1, 1', 2, 3\}, b = \{2,1,3,1\}$ and $B = 5$
- $f(\{1\}) = f(\{1'\}) = 10, \quad f(\{2\}) = 5, \quad f(\{3\}) = 2$
- $f(\{1,1'\}) = 10, f(\{1', 2\}) = 15, f(\{1', 3\}) = 12$
- $f(\{1', 2,3\}) = 16$

Example

Add feasible element $e = \text{argmax}_{e \in N \setminus S,\, \boldsymbol{b}(\boldsymbol{S}) + \boldsymbol{b_e} \leq \boldsymbol{B}} \frac{\boldsymbol{\Delta f(e|S)}}{\boldsymbol{b_e}}$

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- Step 1 of Greedy?

Example

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- Step 1 of Greedy: Add element $1'$
- Step 2?

Example

- $N = \{1, 1', 2, 3\}, b = \{2,1,3,1\}$ and $B = 5$
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- Step 1 of Greedy: Add element $1'$
- Step 2: Add element 3

$$\frac{\Delta f(2\,|\{1'\})}{b_2} = \frac{5}{3}, \qquad \Delta f(1|\{1'\}) = 0, \qquad \frac{\Delta f(3|\{1'\})}{b_3} = 2$$

Example

- $N = \{1, 1', 2, 3\}, b = \{2,1,3,1\}$ and $B = 5$
- $f(\{1\}) = f(\{1'\}) = 10, \quad f(\{2\}) = 5, \quad f(\{3\}) = 2$
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- Step 1 of Greedy: Add element $1'$
- Step 2: Add element 3
- Step 3: Add element 2

$$\frac{\Delta f(2|\{1', 3\})}{b_2} = \frac{4}{3}, \qquad \Delta f(1|\{1', 3\}) = 0 \quad (\text{why?})$$

Approximation Guarantee

- Greedy “value-per-unit-budget” + modifications is $\left(1 - \frac{1}{e}\right)$ approximate
  - Also applies to probability maximization with budget constraint

- No algorithm has better approximation guarantee

Key Take Aways

- Greedy method works well for submodular maximization
  - Best possible approximation guarantee – a metric for evaluating practical heuristics
  - Known to have near-optimal performance in practice

- Probability maximization is a special case of submodular maximization
  - So Greedy works well for probability maximization

- **Without Submodularity: Greedy may not perform well**
  - Important to figure out if the objective is submodular!

Next: Maximize Expected Revenue

1. Maximize probability that something is chosen

$$\max_{S \subseteq [n]} \phi(S)$$

2. Maximize expected revenue from choice

$$\max_{S \subseteq [n]} \sum_{i \in S} r_i \, \phi(i, S)$$

Objectives are identical when prices are identical

Is Revenue Function Submodular?

$$R(S) = \sum_{i \in S} r_i \, \phi(i, S)$$

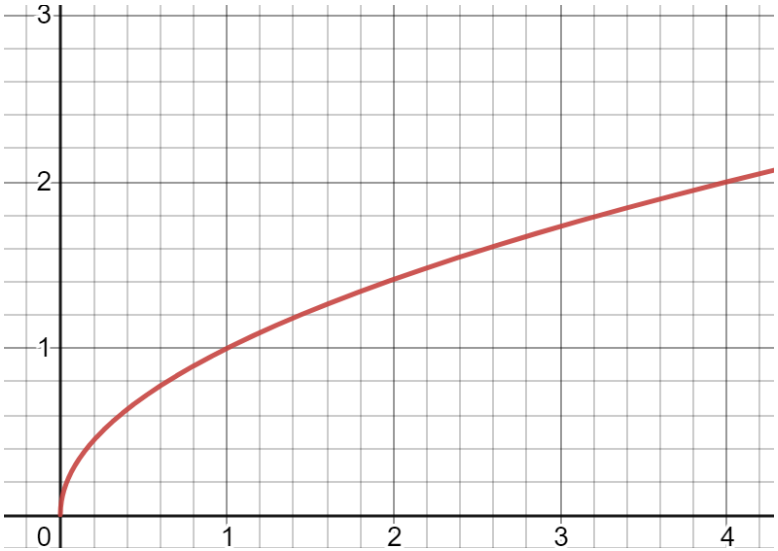
How to Prove Submodularity of a Function?

A Detour

Submodularity of $f(S) = g(|S|)$

Claim: $f(S)$ is submodular for concave functions $g: \mathbb{R} \to \mathbb{R}$

Example of concave function: $g(x) = \sqrt{x}$



Submodularity of $f(S) = g(|S|)$

Claim: $f(S)$ is submodular for concave functions $g: \mathbb{R} \to \mathbb{R}$

Proof:

- Consider arbitrary sets A, B such that $\mathrm{B} \subseteq A$
- Let i be some element in $N \backslash \mathrm{A}$

$$f(A \cup \{i\}) - f(A) = g(|A| + 1) - g(|A|)$$
$$f(B \cup \{i\}) - f(B) = g(|B| + 1) - g(|B|)$$

Submodularity of $f(S) = g(|S|)$

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- Concavity implies that the slope (derivative function $g'(x)$) is decreasing. Thus,

$$g(x + 1) - g(x) \leq g(y + 1) - g(y) \quad \forall\, x \geq y$$

Is this function Submodular?

$$f(S) = \frac{\sum_{i \in S} v_i}{1 + \sum_{i \in S} v_i} \qquad (v_i \geq 0 \; \forall i \in N)$$

- Thoughts?

Is this function Submodular?

$$f(S) = \frac{\sum_{i \in S} v_i}{1 + \sum_{i \in S} v_i} \qquad (v_i \geq 0 \; \forall i \in N)$$

1. Either show that,
   $f(S \cup \{i\}) - f(S) \leq f(S' \cup \{i\}) - f(S') \quad \forall S' \subseteq S, i \in N \backslash S$

2. Or give an example where the inequality does not hold

“Rewriting” the gain of a new element

$$f(S \cup \{e\}) = \frac{v_e + \sum_{i \in S} v_i}{1 + v_e + \sum_{i \in S} v_i}$$

$$= \frac{v_e + \frac{\sum_{i \in S} v_i}{1 + \sum_{i \in S} v_i} (1 + \sum_{i \in S} v_i)}{1 + v_e + \sum_{i \in S} v_i}$$

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$$= \frac{v_e + f(S)(1 + \sum_{i \in S} v_i)}{1 + v_e + \sum_{i \in S} v_i}$$

“Rewriting” the gain of a new element

$$f(S \cup \{e\}) - f(S) = \frac{v_e + \sum_{i \in S} v_i}{1 + v_e + \sum_{i \in S} v_i} - f(S)$$

$$= \frac{v_e + \frac{\sum_{i \in S} v_i}{1 + \sum_{i \in S} v_i}(1 + \sum_{i \in S} v_i)}{1 + v_e + \sum_{i \in S} v_i} - f(S)$$

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$$= \frac{v_e(1 - f(S))}{1 + v_e + \sum_{i \in S} v_i}$$

Is the function Submodular?

$$f(S \cup \{e\}) - f(S) = \frac{v_e\big(1 - f(S)\big)}{1 + v_e + \sum_{i \in S} v_i}$$

$$f(S' \cup \{e\}) - f(S') = \frac{v_e\big(1 - f(S')\big)}{1 + v_e + \sum_{i \in S'} v_i}$$

$$\text{Is } \frac{v_e\big(1-f(S)\big)}{1+v_e+\sum_{i \in S} v_i} \leq \frac{v_e\Big(1-f\big(S'\big)\Big)}{1+v_e+\sum_{i \in S'} v_i} \quad \forall S' \subseteq S \, ?$$

Is the function Submodular?

$$f(S \cup \{e\}) - f(S) = \frac{v_e\left(1 - f(S)\right)}{1 + v_e + \sum_{i \in S} v_i} \qquad \leq \frac{v_e\left(1 - f(S')\right)}{1 + v_e + \sum_{i \in S'} v_i} \; ?$$

$$\leq \frac{v_e\left(1 - f(S')\right)}{1 + v_e + \sum_{i \in S} v_i} \qquad \forall S' \subseteq S$$

$f(S') \leq f(S)$ because $\frac{x}{1+x}$ is an increasing function for $x \geq 0$

Is the function Submodular? – Yes

$$f(S \cup \{e\}) - f(S) = \frac{v_e\left(1 - f(S)\right)}{1 + v_e + \sum_{i \in S} v_i} \qquad \leq \frac{v_e\left(1-f(S')\right)}{1+v_e+\sum_{i \in S'} v_i} \; ?$$

$$\leq \frac{v_e\left(1-f(S')\right)}{1+v_e+\sum_{i \in S} v_i}$$

$$\leq \frac{v_e\left(1-f(S')\right)}{1+v_e+\sum_{i \in S'} v_i} = f(S' \cup \{e\}) - f(S')$$

$$\forall S' \subseteq S$$

Proof Summary

- Let $V=\sum_{i\in S} v_i$, so $f(S)=V/(1+V)$

- Gain from adding e: $v_e(1-f(S))/(1+v_e+V)$
  - Bigger $S \rightarrow f(S)$ is larger, so $1-f(S)$ is smaller
  - Bigger $S \rightarrow$ the denominator is larger
  - Both make the gain smaller → submodular

Is Revenue Function Submodular?

$$R(S) = \sum_{i \in S} r_i \, \phi(i, S)$$

- **Somewhat easier question:** Is the function monotone?

Monotonicity?

$$\max_{S \subseteq [n]} R(S)$$

where $R(S) = \sum_{i \in S} r_i \, \phi(i, S)$

- Revenue is not monotone
  - 2 products, RCM with $\lambda_{1\,2\,0} = \lambda_{2\,0\,1} = 0.5$
  - $r_1 = 1, r_2 = 2$
  - $R(\{2\}) = 2 > 1.5 = R(\{1,2\})$
  - This effect is called “cannibalization”

Revenue is Not Submodular

$$\max_{S \subseteq [n]} R(S)$$

where $R(S) = \sum_{i \in S} r_i \, \phi(i, S)$

- 3 products, RCM with $\lambda_{3\,1\,2\,0} = \lambda_{2\,3\,0\,1} = 0.5$
- $r_1 = 1, r_2 = 2, r_3 = 2$

- $R(\{3,1,2\}) - R(\{1,2\}) = 2 - 1.5 \quad > \quad 0 = R(\{3,2\}) - R(\{2\})$
  - Submodularity requires: $R(A \cup \{3\}) - R(A) \leq R(B \cup \{3\}) - R(B)$

Absence of Structure

$$\max_{S \subseteq [n]} R(S)$$

where $R(S) = \sum_{i \in S} r_i \, \phi(i, S)$

- Function is **not monotone, nor submodular**
- Problem is **very hard to even approximate**
  - Even without any constraints!

The Modeling Dilemma

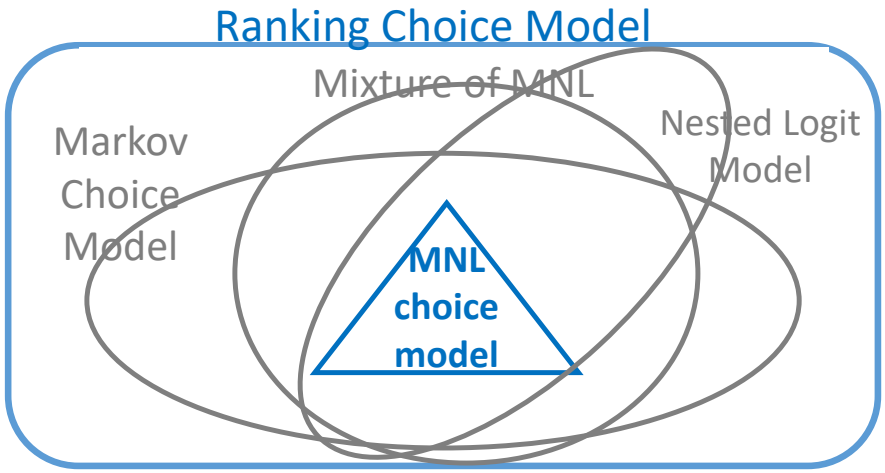
- Real problems are hard to model accurately
- A sophisticated model (such as RCM/RUM) may be more accurate
- Typically, accuracy at the cost of other aspects, such as optimization

Do we need RCM?

- It is very expressive
- But also very hard to **learn** and **optimize**
- Practitioners often prefer simple models
  - We will also see real life application later

Multinomial Logit Model (MNL)

A very simple choice model



Multinomial Logit Choice (MNL)

v_j: weight of product $j \in [n]$

v_0: weight of outside option

$$\phi(i, S) = \frac{v_i}{v_0 + \sum_{j \in S} v_j}$$

- Examples: $\phi(\{1\}) = \frac{v_1}{v_0 + v_1}$, $\phi(\{1\}, \{1,2\}) = \frac{v_1}{v_0 + v_1 + v_2}$

Multinomial Logit Choice (MNL)

$$\phi(i,S) = \frac{v_i}{v_0 + \sum_{j \in S} v_j}$$

- **Optimization and learning are tractable (to be shown)**
- **Often the go-to model, but has clear limitations**